



Practice Exercises for Prompt Engineering

1. Persona Pattern

Exercise Title

AI Career Mentor

Objective

Learn how assigning a role changes the quality of AI responses.

Scenario / Problem Statement

You are a first-year Computer Science student seeking guidance about future career paths.

Student Task

Create a prompt that makes the AI act as a senior software engineer and provide career advice.

Example Solution Prompt

Act as a Senior Software Engineer with 15 years of industry experience.

Guide a first-year Computer Science student on how to become a successful Full-Stack Developer.

Include:

- Skills to learn
- Recommended technologies
- Learning roadmap
- Common mistakes to avoid

Use beginner-friendly language.

Skills Practiced

- Role assignment
- Context setting

- Prompt specificity

Difficulty Level

Beginner

2. Audience Persona Pattern

Exercise Title

One Topic, Three Audiences

Objective

Learn how content changes based on the target audience.

Scenario / Problem Statement

A company wants explanations of Machine Learning for different audiences.

Student Task

Generate explanations for multiple audience types.

Example Solution Prompt

Explain Machine Learning to:

1. A 12-year-old student
2. A BS Computer Science student
3. A CEO of a technology company

Adjust language, examples, and depth appropriately.

Skills Practiced

- Audience adaptation
- Communication skills
- Content customization

Difficulty Level

Beginner

3. Flipped Interaction Pattern

Exercise Title

Programming Language Advisor

Objective

Learn to gather information before generating recommendations.

Scenario / Problem Statement

A student wants to learn a programming language but has not provided enough details.

Student Task

Create a prompt that forces AI to ask questions first.

Example Solution Prompt

Help me choose a programming language.

Before giving recommendations:

- Ask about my goals
- Ask about my experience level
- Ask about my interests
- Ask about my career plans

Only provide recommendations after gathering this information.

Skills Practiced

- Interactive prompting
- Requirement gathering
- Personalized responses

Difficulty Level

Beginner

4. Question Refinement Pattern

Exercise Title

Research Question Optimizer

Objective

Learn to improve vague questions before answering.

Scenario / Problem Statement

Students often ask unclear research questions.

Student Task

Build a prompt that rewrites questions before answering.

Example Solution Prompt

Whenever I ask a question:

1. Rewrite it to make it clearer and more specific.
2. Explain why the improved version is better.
3. Ask whether I want to use the improved version.
4. Only then answer the question.

Skills Practiced

- Prompt improvement
- Clarity enhancement
- Research preparation

Difficulty Level

Beginner

5. Cognitive Verifier Pattern

Exercise Title

Code Logic Checker

Objective

Learn how to verify reasoning before producing answers.

Scenario / Problem Statement

A student receives conflicting advice about algorithm complexity.

Student Task

Create a verification-based prompt.

Example Solution Prompt

Before answering:

1. Identify assumptions.
2. Verify logical reasoning.
3. Check for errors.
4. Highlight uncertainties.
5. Then provide the final answer.

Question:

Is Binary Search faster than Linear Search in all situations?

Skills Practiced

- Critical thinking
- Validation
- Logical reasoning

Difficulty Level

Intermediate

6. Outline Expansion Pattern

Exercise Title

Learn Web Development Roadmap

Objective

Create structured content from an outline.

Scenario / Problem Statement

You want organized study notes for Web Development.

Student Task

Generate an outline and expand every section.

Example Solution Prompt

Create a complete roadmap for learning Web Development.

Step 1:

Generate an outline.

Step 2:

Expand every section with:

- Explanation
- Examples
- Learning resources
- Mini project ideas

Skills Practiced

- Content structuring
- Documentation
- Learning roadmap creation

Difficulty Level

Beginner

7. Fact-Checking Techniques

Exercise Title

Technology Myth Detector

Objective

Verify technology-related claims.

Scenario / Problem Statement

You encounter a statement online.

Student Task

Verify whether the statement is accurate.

Example Solution Prompt

Fact-check the following statement:

"GitHub and Git are exactly the same thing."

Tasks:

- Identify factual claims
- Verify accuracy
- Explain inaccuracies
- Provide corrected information

Skills Practiced

- Research
- Verification
- Critical evaluation

Difficulty Level

Intermediate

8. Few-Shot Prompting

Exercise Title

Technology Classifier

Objective

Teach AI using examples.

Scenario / Problem Statement

You need a system that classifies technology tools.

Student Task

Provide examples and ask AI to continue the pattern.

Example Solution Prompt

Example 1:

Input: Python

Output: Programming Language

Example 2:

Input: PostgreSQL

Output: Database

Example 3:

Input: React

Output: Frontend Framework

Now classify:

Input: Docker

Skills Practiced

- Pattern learning
- Example-driven prompting
- Classification

Difficulty Level

Beginner

9. Prompt Design Best Practices

Exercise Title

Professional LinkedIn Post Creator

Objective

Apply prompt writing best practices.

Scenario / Problem Statement

You completed a Prompt Engineering course and want a LinkedIn post.

Student Task

Write a detailed prompt with clear requirements.

Example Solution Prompt

Write a LinkedIn post announcing my Prompt Engineering certification.

Requirements:

- Professional tone
- 180 words maximum
- Mention key skills learned
- Include GitHub portfolio plans
- End with a networking call-to-action

Skills Practiced

- Prompt structure
- Requirements specification
- Content creation

Difficulty Level

Beginner

10. Structured Prompt Development

Exercise Title

LinkedIn Profile Reviewer

Objective

Use a structured prompt framework.

Scenario / Problem Statement

You want feedback on your LinkedIn profile.

Student Task

Follow Role-Task-Context-Requirements-Output format.

Example Solution Prompt

Role:

Career Coach

Task:

Review my LinkedIn profile summary.

Context:

I am a BS Computer Science student.

Requirements:

Identify strengths and weaknesses.

Output Format:

Bullet points with recommendations.

Skills Practiced

- Prompt frameworks
- Organization
- Professional communication

Difficulty Level

Beginner

11. Prompt Evaluation and Improvement Strategies

Exercise Title

Prompt Quality Auditor

Objective

Evaluate prompt effectiveness.

Scenario / Problem Statement

A prompt is producing poor outputs.

Student Task

Analyze and improve it.

Example Solution Prompt

Evaluate this prompt:

"Teach me programming."

Rate:

- Clarity
- Specificity
- Effectiveness

Score each from 1–10.

Then rewrite the prompt to improve performance.

Skills Practiced

- Prompt evaluation
- Optimization
- Quality assurance

Difficulty Level

Intermediate

12. AI-Assisted Problem Solving Techniques

Exercise Title

Build a Portfolio Plan

Objective

Use AI as a planning partner.

Scenario / Problem Statement

You want to build a professional GitHub portfolio.

Student Task

Ask AI to break the goal into manageable tasks.

Example Solution Prompt

I want to build a professional GitHub portfolio.

Break the goal into:

1. Required skills
2. Portfolio projects
3. Learning roadmap
4. Weekly milestones
5. Resources

Skills Practiced

- Planning
- Goal decomposition
- Strategic thinking

Difficulty Level

Beginner

13. Creative Prompting Methods

Exercise Title

AI Project Brainstormer

Objective

Generate innovative ideas.

Scenario / Problem Statement

You need unique GitHub projects.

Student Task

Use AI for brainstorming.

Example Solution Prompt

Act as an innovation consultant.

Generate 20 unique project ideas that combine:

- Prompt Engineering
- Artificial Intelligence
- GitHub Portfolio Development

Include difficulty and learning outcomes.

Skills Practiced

- Creativity
- Brainstorming
- Innovation

Difficulty Level

Beginner

14. Analytical Prompting Methods

Exercise Title

Programming Language Comparison

Objective

Perform structured analysis.

Scenario / Problem Statement

A student must choose a language for future projects.

Student Task

Compare multiple technologies.

Example Solution Prompt

Compare Python, Java, and C++.

Analyze:

- Learning difficulty
- Job opportunities
- Community support
- Project suitability

Present findings in a table.

Skills Practiced

- Comparative analysis
- Decision support
- Technical evaluation

Difficulty Level

Intermediate

15. Human-AI Collaboration Approaches

Exercise Title

Project Idea Enhancement

Objective

Collaborate with AI to improve ideas.

Scenario / Problem Statement

You have a basic project idea but need refinement.

Student Task

Ask AI to improve and critique your idea.

Example Solution Prompt

I will provide a project idea.

Your tasks:

1. Improve the idea

2. Identify weaknesses
3. Suggest additional features
4. Create an implementation roadmap

Wait for my idea first.

Skills Practiced

- Collaboration
- Project planning
- Critical evaluation

Difficulty Level

16. Chain-of-Thought (COT) Prompting

Exercise Title

Debug the Login System

Objective

Learn how step-by-step reasoning improves problem-solving.

Scenario / Problem Statement

A login system is rejecting valid users. You need AI to analyze the issue logically before suggesting solutions.

Student Task

Create a prompt that forces AI to reason through the problem step-by-step.

Example Solution Prompt

A web application login system is rejecting valid users.

Think through the problem step-by-step.

Analyze:

1. Possible causes
2. Authentication process
3. Database issues
4. Password validation errors
5. Recommended fixes

Provide the final diagnosis and solution.

Skills Practiced

- Logical reasoning
- Debugging
- Problem-solving

Difficulty Level

Intermediate

17. Zero-Shot Prompting

Exercise Title

Quick Technical Summary

Objective

Learn how to perform tasks without providing examples.

Scenario / Problem Statement

You found a technical article and need a quick summary.

Student Task

Write a direct prompt without examples.

Example Solution Prompt

Summarize the following article in 150 words.

Focus on:

- Main idea
- Key findings
- Practical applications

Use simple language suitable for Computer Science students.

Skills Practiced

- Direct prompting
- Summarization
- Information extraction

Difficulty Level

Beginner

18. Role-Based Prompting

Exercise Title

Cybersecurity Awareness Trainer

Objective

Learn how professional roles influence responses.

Scenario / Problem Statement

Students need cybersecurity awareness training.

Student Task

Assign a professional role to AI.

Example Solution Prompt

Act as a Cybersecurity Expert.

Explain:

1. Phishing attacks
2. Social engineering
3. Password security
4. Safe browsing habits

Provide real-world examples relevant to university students.

Skills Practiced

- Role assignment
- Security awareness

- Educational content creation

Difficulty Level

Beginner

19. Meta Prompting

Exercise Title

Prompt Generator Assistant

Objective

Use AI to create better prompts.

Scenario / Problem Statement

You need a prompt that helps students learn GitHub efficiently.

Student Task

Ask AI to generate an optimized prompt.

Example Solution Prompt

Create an advanced prompt that will help a first-year Computer Science student learn Git and GitHub effectively.

The generated prompt should:

- Include a learning roadmap
- Include practical exercises
- Include project ideas
- Include milestones

Skills Practiced

- Prompt optimization
- Prompt engineering
- Workflow design

Difficulty Level

Intermediate

20. Iterative Prompting

Exercise Title

Improve a LinkedIn Post

Objective

Learn how repeated refinement improves outputs.

Scenario / Problem Statement

Your first LinkedIn draft looks average.

Student Task

Improve the output through multiple iterations.

Example Solution Prompt

Round 1:

Write a LinkedIn post about completing a Prompt Engineering course.

Round 2:

Make the tone more professional.

Round 3:

Add achievements and learning outcomes.

Round 4:

Improve the conclusion and call-to-action.

Skills Practiced

- Content refinement
- Iterative improvement
- Professional writing

Difficulty Level

Beginner

21. Constraint-Based Prompting

Exercise Title

100-Word Git Explanation

Objective

Learn how constraints affect outputs.

Scenario / Problem Statement

You must explain Git within strict limitations.

Student Task

Create a prompt with multiple constraints.

Example Solution Prompt

Explain Git.

Constraints:

- Maximum 100 words
- Simple English
- Include one real-world example
- No technical jargon
- Use bullet points

Skills Practiced

- Constraint management
- Precision writing
- Output control

Difficulty Level

Beginner

22. Multi-Step Prompting

Exercise Title

Create a Student Portfolio

Objective

Break large tasks into smaller stages.

Scenario / Problem Statement

You want to build a complete GitHub portfolio.

Student Task

Guide AI through multiple sequential tasks.

Example Solution Prompt

Help me create a GitHub portfolio.

Step 1:

Identify portfolio sections.

Step 2:

Suggest projects.

Step 3:

Design repository structure.

Step 4:

Create README content.

Step 5:

Suggest improvements.

Skills Practiced

- Task decomposition
- Project planning
- Workflow design

Difficulty Level

Intermediate

23. Reflection Prompting

Exercise Title

Self-Review Blog Writer

Objective

Learn how AI can critique its own work.

Scenario / Problem Statement

You wrote a blog post but want quality improvements.

Student Task

Ask AI to review and improve its response.

Example Solution Prompt

Write a blog post about GitHub for beginners.

After writing:

1. Review your work.
2. Identify three weaknesses.
3. Suggest improvements.
4. Rewrite the improved version.

Skills Practiced

- Self-evaluation
- Quality improvement
- Content optimization

Difficulty Level

Intermediate

24. Self-Consistency Prompting

Exercise Title

Find the Best Learning Strategy

Objective

Generate multiple reasoning paths and compare them.

Scenario / Problem Statement

A student wants to learn programming efficiently.

Student Task

Request multiple approaches before choosing one.

Example Solution Prompt

A student wants to learn programming in six months.

Generate three different learning strategies.

For each strategy:

- Explain the approach
- Analyze strengths
- Analyze weaknesses

Compare the results and recommend the most reliable strategy.

Skills Practiced

- Decision-making
- Comparative reasoning
- Evaluation

Difficulty Level

Intermediate

25. Tree-of-Thought Prompting

Exercise Title

Choose an Online Career Path

Objective

Explore multiple solutions before selecting the best one.

Scenario / Problem Statement

A Computer Science student wants to earn money online.

Student Task

Generate multiple career paths and evaluate each.

Example Solution Prompt

I am a Computer Science student who wants to earn online.

Explore three possible paths:

1. Freelancing
2. Remote internships
3. Content creation

For each path:

- Requirements
- Advantages
- Disadvantages
- Income potential

Recommend the best option.

Skills Practiced

- Strategic thinking
- Decision analysis
- Career planning

Difficulty Level

Intermediate

26. Re Act (Reason + Act) Prompting

Exercise Title

GitHub Portfolio Auditor

Objective

Combine reasoning with actionable recommendations.

Scenario / Problem Statement

You want to improve your GitHub portfolio.

Student Task

Ask AI to analyze and then suggest actions.

Example Solution Prompt

Analyze my GitHub portfolio.

First:

Reason about its strengths and weaknesses.

Then:

Provide specific actions to improve:

- Repository organization
- README quality
- Documentation
- Portfolio projects
- Professional appearance

Skills Practiced

- Analysis
- Action planning
- Portfolio improvement

Difficulty Level

Intermediate

27. Socratic Prompting

Exercise Title

Discover OOP Yourself

Objective

Learn through guided questioning.

Scenario / Problem Statement

You want to understand Object-Oriented Programming deeply.

Student Task

Force AI to teach using questions only.

Example Solution Prompt

Help me learn Object-Oriented Programming.

Do not explain directly.

Instead:

- Ask guiding questions
- Build concepts gradually
- Help me discover the answers myself
- Correct misunderstandings when necessary

Skills Practiced

- Critical thinking
- Active learning
- Concept discovery

Difficulty Level

Intermediate

28. Reverse Prompting

Exercise Title

Become a Prompt Engineer in One Year

Objective

Work backward from a desired outcome.

Scenario / Problem Statement

You want to become job-ready as a Prompt Engineer.

Student Task

Start from the goal and plan backward.

Example Solution Prompt

I want to become a professional Prompt Engineer within one year.

Work backward from this goal.

Create:

- Monthly milestones
- Skills to acquire
- Projects to complete
- Portfolio requirements
- Learning resources

Skills Practiced

- Goal planning
- Strategic thinking
- Career development

Difficulty Level

Intermediate

29. Comparative Prompting

Exercise Title

Choose Your First Programming Language

Objective

Compare alternatives systematically.

Scenario / Problem Statement

A beginner is confused between Python, Java, and C++.

Student Task

Create a structured comparison.

Example Solution Prompt

Compare Python, Java, and C++.

Evaluate:

- Learning difficulty
- Community support
- Career opportunities
- Project development
- Industry demand

Present the results in a comparison table and recommend the best option for beginners.

Skills Practiced

- Comparative analysis
- Decision support
- Technical evaluation

Difficulty Level

Beginner

30. Scenario-Based Prompting

Exercise Title

Mock Software Developer Interview

Objective

Learn through realistic simulations.

Scenario / Problem Statement

You have an upcoming internship interview and need practice.

Student Task

Create a realistic interview simulation.

Example Solution Prompt

Act as a hiring manager interviewing me for a Junior Software Developer internship.

Requirements:

- Ask one question at a time
- Mix technical and behavioral questions
- Provide feedback after each answer
- Rate my performance
- Suggest improvements

Make the interview realistic.

Skills Practiced

- Interview preparation
- Communication
- Real-world simulation

Difficulty Level

Intermediate